

Magnetic Field Scanner - Future Ideas & Roadmap

This document captures brainstormed ideas for increasing the usability, capability, and professional feel of the MFS wand for Phase 5 and beyond.

1. The Screen & UI Experience

- **Scrolling "Seismograph" Strip Chart:** Add a 4th LVGL tile containing an `lv_chart`. As the user walks over a buried pipe, it will draw the "bell curve" of the magnetic anomaly in real-time. This visually pinpoints the exact center of the pipe much better than reading a rapidly changing integer.
- **Target Depth Estimation:** With the addition of the 3rd (NEAR) sensor at 8 inches, implement a depth-inversion algorithm to calculate the physical depth of the target and display a live "Estimated Depth: XX inches" readout.

2. Connectivity & Web Portal

- **Over-The-Air (OTA) Updates:** Add an HTTP endpoint to `web_server.cpp` utilizing `Update.h` to allow uploading .bin compiled firmware directly from a phone or laptop browser, avoiding USB tethering.
- **QR Code Pairing:** Have the LVGL screen generate and display a QR code containing the wand's IP address. Users can scan it with their phone camera to instantly open the Web Portal without typing numbers.
- **Live WebSocket Streaming:** Implement WebSockets to stream the 400Hz telemetry live to a connected browser. This enables viewing high-resolution charts on a laptop in real-time during a scan.

3. Advanced Physics & Math

- **Kalman Filter for Auto-Tare:** Replace the simple Exponential Moving Average (EMA) auto-tare logic with a 1D Kalman filter to better isolate slow temperature drift from the slow approach of a deep magnetic target.
- **GPS Mapping:** Integrate a standard NMEA GPS module (using the freed IO43 and IO44 pins) to attach Lat/Lon coordinates to every row in the CSV log. This allows for post-processing the CSV into a visual heat-map overlay on Google Earth or QGIS.

4. Deep Search Mode (Software Oversampling)

Concept: Fight the inverse-square law to double the detection depth of the wand without requiring a hardware spin.

- **Physics:** Because magnetic fields decay rapidly, doubling the detection range requires reducing the noise floor by a factor of 4.
- **Implementation:** Apply a Rolling Software Average (Oversampling) to the raw hardware values before calculating the gradient.
- **Benefit over Hardware CC:** Increasing the hardware Cycle Count (e.g., to CC=6400) would reduce the noise, but it would drop the UI refresh rate to 3 FPS (massive stutter). A software rolling average over 16 samples keeps the FreeRTOS loop and UI running at a buttery-smooth 50 FPS, it just introduces a slight phase delay (~300ms smear) requiring the user to sweep slower.
- **Proposed UI:** Instead of cluttering the Cycle Count button with 11 different mathematical combinations (which causes decision paralysis), split the UI into two logical toggles: 1. **Hardware Speed:** [FAST: 200] -> [NORMAL: 400] -> [MAX: 800] 2. **Deep Search (Averaging):** [OFF: 1x] -> [DEEP: 8x] -> [ULTRA: 16x]

5. Hardware Power Isolation (LDO Separation)

Concept: The ESP32-S3 and the LCD backlight create significant electrical noise on the 3.3V power rail, which ripples into the RM3100 analog measurement.

- **Implementation:** On the next PCB revision, split the analog and digital power planes. Provide a dedicated, ultra-low-noise LDO voltage regulator exclusively for the RM3100 I2C bus.
- **Benefit:** Shaves an estimated 10-20% off the noise floor with zero impact on UI speed or latency.

6. Physical Characterization Jig Improvements

Concept: Hand-sweeping a powerful magnet toward the sensor introduces temporal inconsistencies and triggers long hardware timeouts (e.g., 480ms) right at the boundary of saturation. This makes mapping the exact spatial geometry of the core collapse very difficult.

- **Implementation:** Construct a 3D-printed or wooden jig utilizing a non-ferrous threaded rod (e.g., brass or nylon). One full rotation of the rod advances the magnet exactly 1mm (or 1/16th inch).
- **Benefit:** Allows the user to precisely step the magnet toward the sensor in absolute $\frac{1}{16}$ increments, waiting for all thermal and temporal transients to settle before recording the log frame. This guarantees perfect spatial mapping of the saturation cliff.
- **Additional Enhancement:** Angle the target (e.g., rebar) at exactly 45° Pitch and 45° Yaw relative to the wand during tests to perfectly split the field across the X, Y, and Z axes. This distributes the magnetic load and theoretically increases the digital integer overflow ceiling by $\sqrt{3}$.